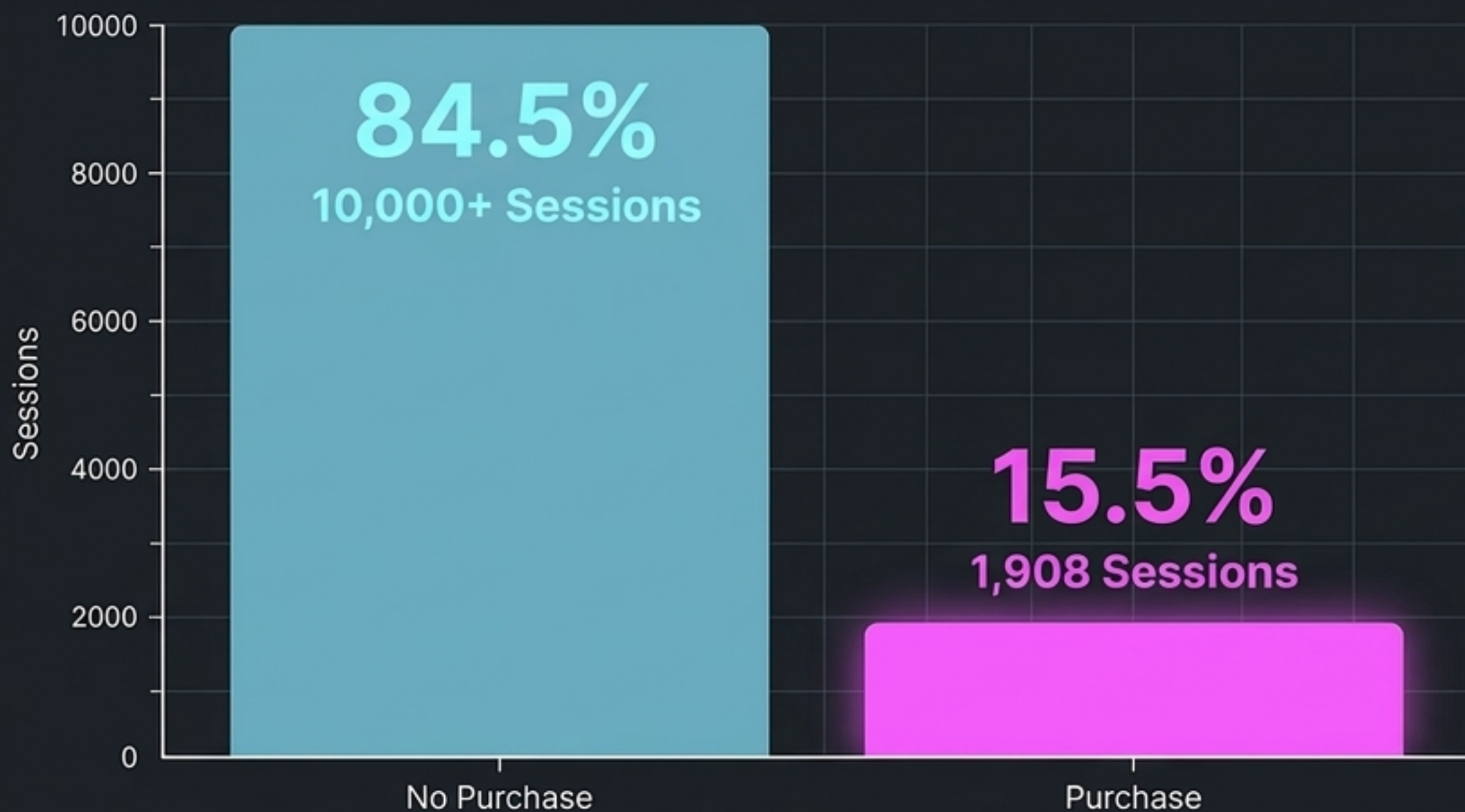


Online Shopper Purchasing Intention

An end-to-end ML classification engine to predict conversions, triage users, and drive timely business interventions.

The Reality of E-Commerce

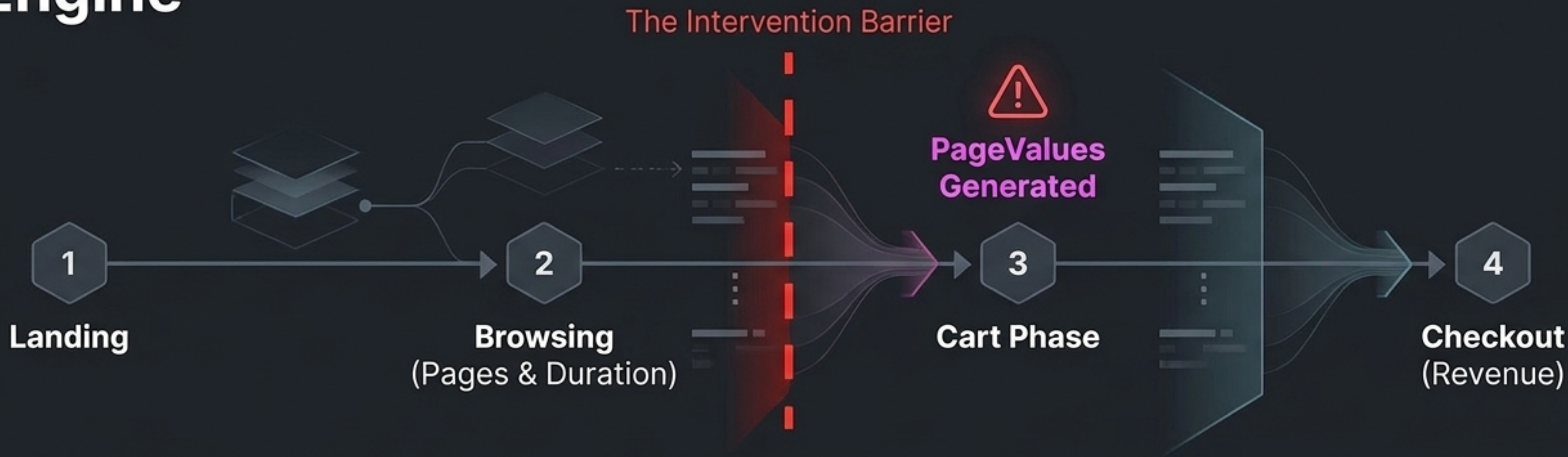


The Accuracy Illusion

A model that simply guesses "no purchase" every time is 84.5% accurate—but generates zero revenue.

To win, the engine must isolate the rare, highly valuable converting sessions without triggering massive false alarms. Success requires **optimizing for precision, recall, and F1 score**.

The Leakage Trap: Engineering a True Predictive Engine



The Target Proxy Problem

We intentionally threw away the dataset's strongest signal: PageValues. This Google Analytics metric implies items are already in a cart. Relying on it tells you a user will buy only after they have already decided to buy.

The Engineering Constraint

True early-intent prediction must operate in the blind spot before checkout. The model relies entirely on 16 ambient behavioral indicators (durations, bounce rates, visit type) available prior to the intervention line.

Model Selection & Pipeline Integrity

Baseline (Always Negative)	Logistic Regression	Random Forest (Selected)
84.5% Accuracy, 0% Recall. Useless for business intervention.	High interpretability. Requires strict feature scaling and imputation.	Captures non-linear behavioral patterns natively. Optimized for Validation Average Precision & F1 score.

```
from sklearn.impute import SimpleImputer
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

page_activity_columns = [
    "Administrative", "Administrative_Duration",
    "Informational", "Informational_Duration",
    "ProductRelated", "ProductRelated_Duration",
]

pca_experiment = make_pipeline(
    SimpleImputer(strategy="median"),
    StandardScaler(),
    PCA()
)

pca_experiment.fit(X_train[page_activity_columns])
```

Zero Data Leakage

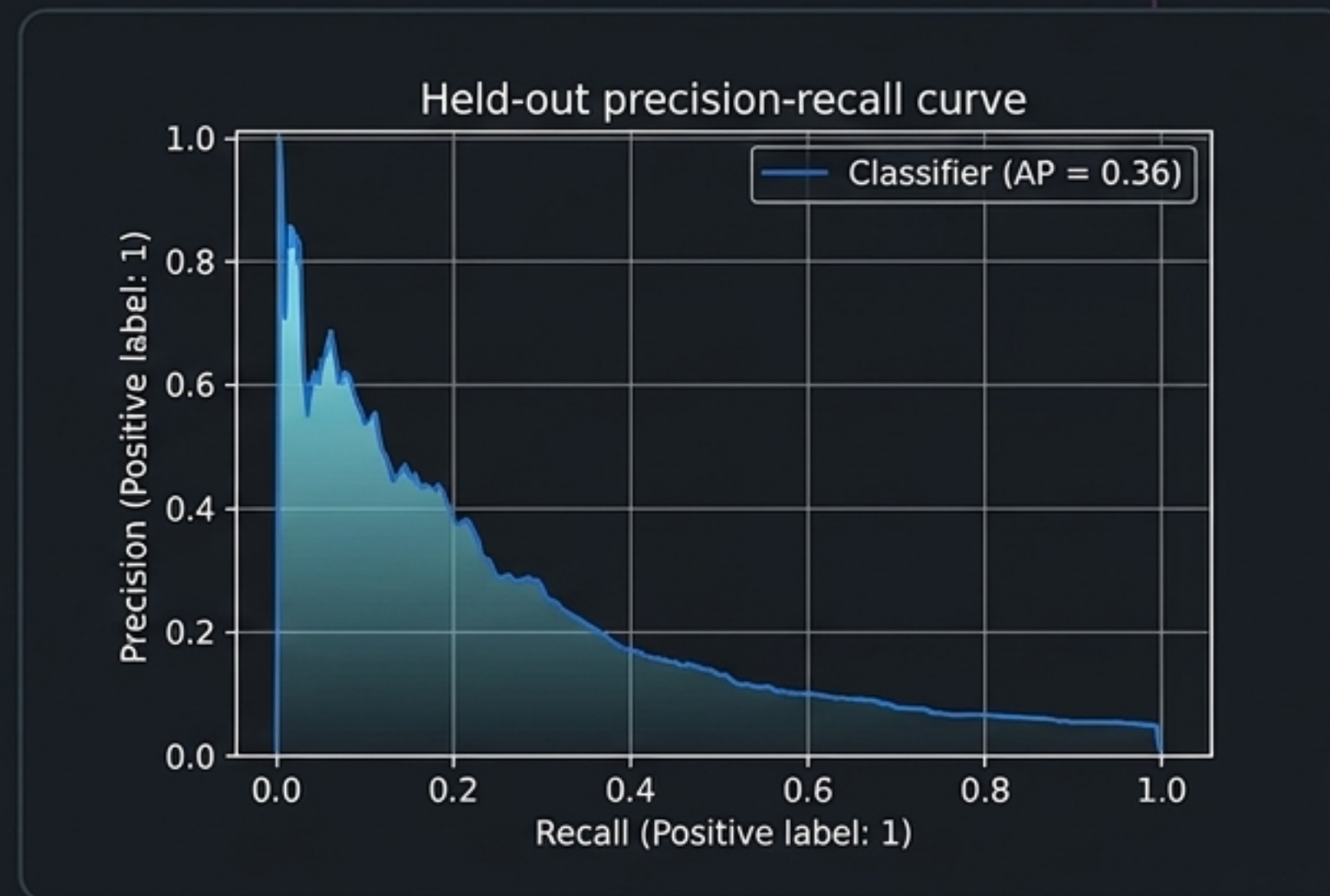
Preprocessing (learned imputation, scaling, categorical encoding) is baked directly into the scikit-learn pipeline and fitted exclusively on the 60% training split to guarantee robust performance on unseen data.

Evaluating Predictive Power



Held-out confusion matrix

True label	No purchase	1533	551
	Purchase	144	238
	Predicted label	No purchase	Purchase

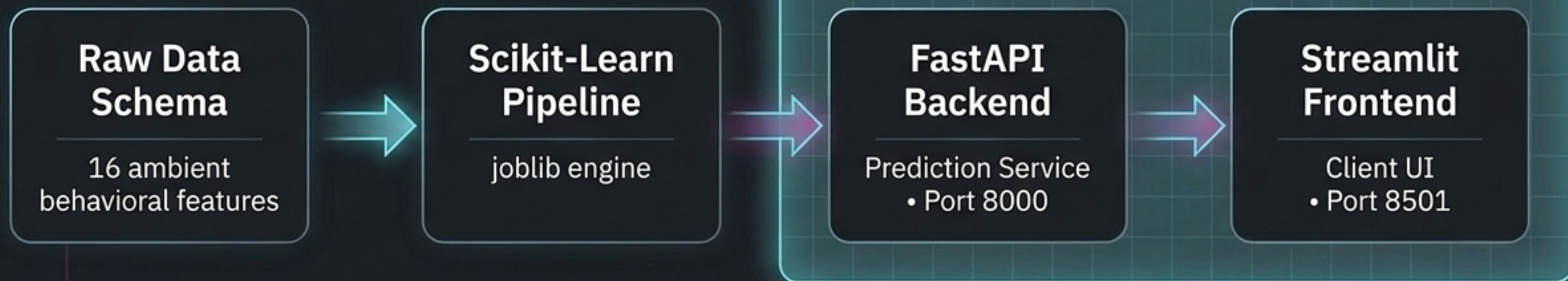


The Flexible Business Dial

The probability threshold (0.319) is not a rigid guess. It is intentionally tuned to maximize Validation F1, acting as a flexible dial that a business can adjust based on their specific marketing budget. Lower the threshold to cast a wider net for cheap interventions, or raise it for expensive discounts.

Full-Stack Architecture: Beyond the Notebook

Tech-Stack Blueprint



Separation of Concerns: The rigorous ML feature contract is enforced by the FastAPI backend, keeping the Streamlit UI lightweight. A true 1-click cloud-deployable **software service**.

Product Feature 1: Live Session Triage

1. Input Variables

Requires exactly 16 behavioral features. Identifiers and target proxies intentionally excluded.

2. System Status

Real-time health check:
Model ready - version 1.0.0.

3. Action Trigger

Instantly gauges intent via a live POST request to the FastAPI endpoint. Built to integrate directly with live storefronts.



Online Shopper Classifier

Analyze real online shopping sessions and predict purchase likelihood using a trained machine learning model. Enter a session manually or upload a CSV to classify shoppers into likelihood based on their observed browsing behavior.

Single session Batch CSV upload

Enter one session

Administrative pages

2

- +

Product related pages

18

- +

Special day proximity

0.00

- +

Administrative duration

25.00

- +

Product related duration

620.50

- +

Month

June

▼

Informational pages

0

- +

Bounce rate

0.018

- +

Visitor type

Returning_Visitor

▼

Informational duration

0.00

- +

Exit rate

0.030

- +

☐ Weekend session

Session context

Operating system code

2

- +

Region code

1

- +

Browser code

2

- +

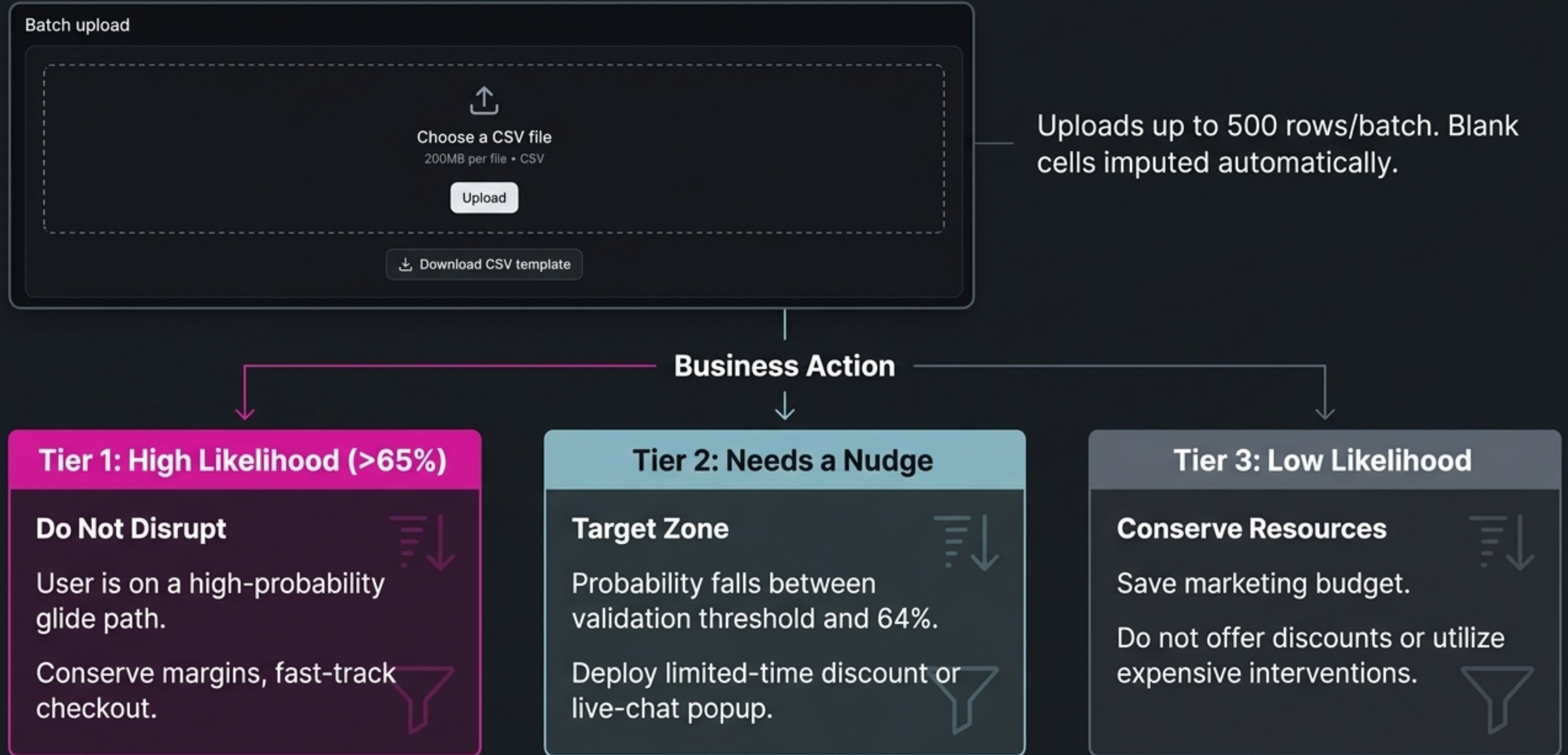
Traffic type code

2

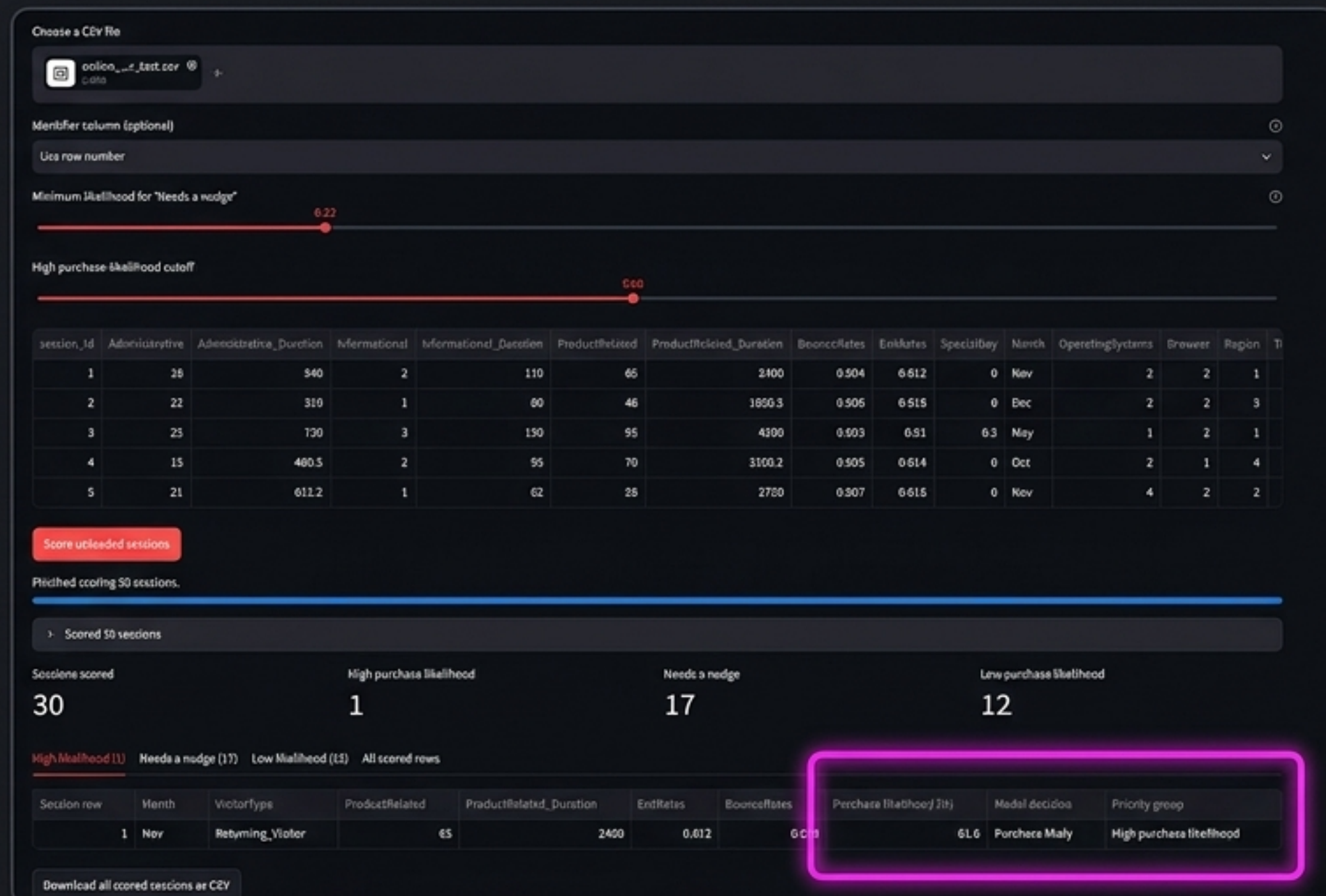
- +

Estimate purchase probability

Product Feature 2: The Triage Framework



Operational Synthesis: Data to Action



- **The Translation Layer:** Bridges the gap between engineering and marketing. Abstract numeric inputs are instantly translated into plain-English directives.
- **The Outcome:** Generates a downloadable, segmented list ready for immediate CRM ingestion, scoring hundreds of sessions in seconds.

Limitations & Trajectory



Current Limitations

Uses historical, aggregated session data.

Non-causal prediction (identifies intent, but doesn't guarantee feature changes cause purchase).



The Next Evolution

Transitioning from batch historical data to real-time Kafka event streams, and linking predictions directly to unique customer Lifetime Value (LTV).



Call to Action

The stack is containerized, fully documented, and ready for technical evaluation today.

```
> docker run --rm -p 8000:8000 -p 8501:8501 online-shopper-intention:1.0
```